

Problem set 7

Information Retrieval Problems

Excercise 1

The words car, auto, best should have the following frequencies in the ten documents Doc1, .., Doc10. Simply assume that all other words in the documents are stop words. Compute the tf-idf weight for each term.

term\doc	Doc1	Doc2	Doc3	Doc4	Doc5	Doc6	Doc7	Doc8	Doc9	Doc10
car	3	0	0	5	12	0	0	2	8	1
auto	8	6	0	12	0	0	9	1	3	10
best	0	1	7	0	1	5	12	0	2	0

Figure 1: Term Frequency Table for Each Document.

Excercise 2

Assume the following documents. Assume that the following terms are stop words: is, and, in, the, was, each, other. Construct an inverted index.

Assume the following documents

doc-id\text	
Doc1	"italy is world champion 2006"
Doc2	"germany and italy played each other in the semifinal"
Doc3	"germany was in the semifinal 2006"
Doc4	"germany won the semifinal in italy 1990"

Figure 2: Documents in a corpus.

Excercise 3

Consider following documents in the corpus.

- Doc 1 breakthrough drug for schizophrenia.
- Doc 2 new schizophrenia drug.
- Doc 3 new approach for treatment of schizophrenia.

- Doc 4 new hopes for schizophrenia patients.

Assume the following documents

doc-id	text
Doc1	"Italy is world champion 2006"
Doc2	"Germany and Italy played each other in the semifinal"
Doc3	"Germany was in the semifinal 2006"
Doc4	"Germany won the semifinal in Italy 1990"

Figure 3: Documents in a corpus.

What are the returned results for these queries:-

- schizophrenia AND drug
- schizophrenia AND NOT drug
- schizophrenia OR drug

Exercise 4

Recommend a query processing order for

(tangerine OR trees) AND (marmalade OR skies) AND (kaleidoscope OR eyes)

given the following postings list sizes:

Term	Postings size
eyes	213312
kaleidoscope	87009
marmalade	107913
skies	271658
tangerine	46653
trees	316812

Table 1: Terms and the corresponding posting list size

Exercise 5

In constructing the index, which step is most expensive/complex?

Exercise 6

Why don't we use a relational database for information retrieval?

Exercise 7

In the BM25/Okapi formula, which term is the ‘inverse document frequency’? Which term is the ‘term frequency’? Where is the length normalization?

Exercise 8

Given the following corpus:

- D1 I love cats cats cats.
- D2 I love cats and I love dogs.
- D3 I love dogs.
- D4 I dont like cats.
- D5 I love puppies.
- D6 I love kittens.

And the following queries:

- **Q1:** cats
- **Q2:** dogs

First, start with standard assumptions: a large corpus ($N=1000$), few relevant documents ($S=3$), and each term occurs in only the documents above. Because the numbers are small relative to N , you can assume that $N-n$ or $N-S$ is approximately N . Also assume $p_i=0.5$; i.e., if a term is relevant, it has a 0.5 probability of occurring in a relevant document. Calculate the Retrieval Status Value for each query and the documents above, and rank the documents.

Exercise 9

How does the size of the index impact search performance? Does a larger index size guarantee a better performance? Why or why not? (Remember, search performance is asking how relevant the returned documents are, not how fast it runs.)

Exercise 10

What is the idf of a term that occurs in every document? Compare this with the use of stop word lists.